

1. Causality & Probability Toolkit

3 correlation cases: X, Y correlated: (i) causality $X \rightarrow Y$; (ii) reverse causality $Y \rightarrow X$; (iii) common factor $C \rightarrow X$ and $C \rightarrow Y$ (confounder). Correlation alone never tells you which.

E/Var/Cov rules: $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$; $\text{Var}(aX + b) = a^2 \text{Var}(X)$; $\text{Cov}(aX + b, Y) = a \text{Cov}(X, Y)$, $\text{Cov}(X, Y + Z) = \text{Cov}(X, Y) + \text{Cov}(X, Z)$; $\text{Var}(X + Y) = \text{Var} X + \text{Var} Y + 2 \text{Cov}(X, Y)$.

Independence: $X \perp Y \Rightarrow \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] \Rightarrow \text{Cov}(X, Y) = 0$. Converse *false*: $\text{Cov} = 0$ (uncorrelated) does not imply independent (nonlinear dependence possible).

LIE: $\mathbb{E}[\mathbb{E}(Y|X)] = \mathbb{E}[Y]$. Fast route to $\mathbb{E}[Y]$ or $\mathbb{E}[g(X)Y]$: find $\mathbb{E}[Y|X]$ first, then average over X ; cross-check against the direct expansion.

LIE worked example: $X \in \{0, 1\}$ with $P(X=1) = 0.3$; $\mathbb{E}[Y|X=1] = 10$, $\mathbb{E}[Y|X=0] = 4$; $\mathbb{E}[Y] = 0.3(10) + 0.7(4) = 3 + 2.8 = 5.8$.

$N(0, 1)$ **moments:** $X \sim N(0, 1)$: $\mathbb{E}[X] = 0$, $\mathbb{E}[X^2] = 1$, $\mathbb{E}[X^3] = 0$ (all odd moments vanish by symmetry of ϕ).

Type I/II & significance: Type I = reject true H_0 (false positive); Type II = fail to reject false H_0 (false negative). Significance level := $P(\text{Type I} | H_0 \text{ true})$, usually 10/5/1% (0.1% legal/medical). Stricter cutoff $\Rightarrow$ Type I $\downarrow$, Type II $\uparrow$; cannot shrink both at once.

RCT logic: Random assignment $\Rightarrow X \perp u$ (full independence) $\Rightarrow \mathbb{E}[u|X] = \mathbb{E}[u] = 0$. Causal effect $\approx \mathbb{E}[Y|\text{treatment}] - \mathbb{E}[Y|\text{control}]$; RCTs often infeasible in economics $\Rightarrow$ need observational-data tools (Ch. 3–).

Case examples: Beer sales vs. A/C sales: confounded by temperature (case ii). Windmill speed vs. wind speed: wind causes windmill (case i, not ii). Ice-cream sales vs. drowning: common factor temperature – regress each on temperature separately, never one on the other.

Conditional mean & population/sample: $\mathbb{E}(Y|X = x)$ = population mean of Y among units with $X = x$; e.g. gender wage gap = $\mathbb{E}(\text{wage}|\text{male}) - \mathbb{E}(\text{wage}|\text{female})$. *Population* = every unit; *sample* = the units actually drawn – keep the two distinct.

COVID ART example: Raise threshold 10 $\rightarrow$ 100 units: Type I $\downarrow$ (fewer healthy people flagged), Type II $\uparrow$ (more infected missed). Lower threshold 10 $\rightarrow$ 1: Type I $\uparrow$, Type II $\downarrow$ – illustrates the trade-off with a concrete knob.

2. The OLS Estimator

Model: $Y_i = \beta_0 + \beta_1 X_i + u_i$, $i = 1, \dots, n$; Y_i dependent var, X_i regressor, u_i = every other factor affecting Y_i .

Three betas: (β_0, β_1) true unknown params; (b_0, b_1) generic candidates inside the minimization; $(\hat{\beta}_0, \hat{\beta}_1)$ the random OLS *estimators* (minimizers). Never conflate the three.

OLS objective: $(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{b_0, b_1} S(b_0, b_1)$, $S(b_0, b_1) = \sum_i [Y_i - (b_0 + b_1 X_i)]^2$ (sum of squared prediction errors).

FOCs: $\frac{\partial S}{\partial b_0} = -2 \sum_i (Y_i - b_0 - b_1 X_i) = 0$; $\frac{\partial S}{\partial b_1} = -2 \sum_i (Y_i - b_0 - b_1 X_i) X_i = 0$.

Normal equations: (4) $\bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X} = 0$; (5) $\frac{1}{n} \sum_i X_i Y_i - \hat{\beta}_0 \bar{X} - \hat{\beta}_1 (\frac{1}{n} \sum_i X_i^2) = 0$. Solve (4) for $\hat{\beta}_0$, substitute into (5), isolate $\hat{\beta}_1$.

$\hat{\beta}_1, \hat{\beta}_0$ **formulas:** $\hat{\beta}_1 = \frac{\frac{1}{n} \sum_i X_i Y_i - \bar{X} \bar{Y}}{\frac{1}{n} \sum_i X_i^2 - \bar{X}^2} = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{s_{XY}}{s_X^2}$; $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$. Needs $\sum_i (X_i - \bar{X})^2 > 0$ (X must vary).

Summation identities: $\sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_i X_i Y_i - n \bar{X} \bar{Y}$; $\sum_i (X_i - \bar{X})^2 = \sum_i X_i^2 - n \bar{X}^2$. Correct $\frac{1}{n}$ form (both sides): $\frac{1}{n} \sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = \frac{1}{n} \sum_i X_i Y_i - \bar{X} \bar{Y}$ – a lone $\frac{1}{n}$ on the RHS only is wrong by a factor of n .

Mean point: OLS line always passes through $(\bar{X}, \bar{Y})$: $\hat{\beta}_0 + \hat{\beta}_1 \bar{X} = \bar{Y}$; recovers $\bar{Y}$ from $\bar{X}$ with no raw data.

Worked $\hat{\beta}_1, \hat{\beta}_0$ example: Given $\bar{X} = 10$, $\bar{Y} = 50$, $s_{XY} = 8$, $s_X^2 = 4$: $\hat{\beta}_1 = 8/4 = 2$; $\hat{\beta}_0 = 50 - 2(10) = 30 \Rightarrow \hat{Y} = 30 + 2X$.

Fitted value & residual: $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$; $\hat{u}_i = Y_i - \hat{Y}_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)$.

Prediction recipe: At $X = x_0$: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x_0$. Change in prediction for ΔX : $\Delta \hat{Y} = \hat{\beta}_1 \Delta X$, or the difference of two direct predictions – both must agree.

Convexity check: Hessian of $S(b_0, b_1)$: leading minor $2n > 0$, $\det H = 4n \sum_i (X_i - \bar{X})^2 > 0 \Rightarrow$ positive definite $\Rightarrow$ the FOC stationary point is the unique global minimum (not just a critical point).

California reference: $n = 420$: TestScore = 698.9 – 2.28 STR. At STR= 20: $\hat{Y} = 653.3$; a 1-student cut in STR $\Rightarrow +2.28$ points.

Interpretation traps: Units of $\hat{\beta}_1 = (\text{units of } Y)/(\text{unit of } X)$. Never quote $\hat{\beta}_0$ as a valid prediction at $X = 0$ outside the observed range of X (extrapolation); $\hat{\beta}_1$ is a *random* estimator, not the fixed β_1 .

CEO salary residual example: $n = 209$: roe= 14.1 $\Rightarrow \widehat{\text{salary}} = 1224.05$; actual salary = 1095 $\Rightarrow \hat{u} = -129.06$ – an individual district/firm can deviate from the “on average” fitted line.

3. Least Squares Assumptions

LSA 1 (Least Squares Assumption 1): $\mathbb{E}[u_i | X_i] = 0$ (zero conditional mean).

LSA 2: (X_i, Y_i) , $i = 1, \dots, n$, are i.i.d.

LSA 3: Large outliers unlikely (informally: finite fourth moments).

$\mathbb{E}[Y|X]$ line: LSA1 $\Rightarrow \mathbb{E}[Y_i | X_i] = \beta_0 + \beta_1 X_i$: at any fixed X , the *average* Y sits exactly on the population line; individual u_i scatter points above/below.

LSA1 $\Rightarrow$ moments, not $\Leftarrow$: $\mathbb{E}[u|X] = 0 \Rightarrow \mathbb{E}[u] = 0$ and $\text{Cov}(X, u) = 0$ (via LIE) – *not conversely*. Counterexample: $X \sim N(0, 1)$, $u = X^2 - 1$: $\mathbb{E}[u] = 0$, $\text{Cov}(X, u) = 0$, yet $\mathbb{E}[u|X] = X^2 - 1 \neq 0$ except at $X = \pm 1$.

Residual identities: $\sum_i \hat{u}_i = 0$ (FOC $_{b_0}$); $\sum_i \hat{u}_i X_i = 0$ (FOC $_{b_1}$); $\sum_i \hat{u}_i \hat{Y}_i = \hat{\beta}_0 \sum_i \hat{u}_i + \hat{\beta}_1 \sum_i \hat{u}_i X_i = 0$. Always true *by construction*, regardless of LSA1–3.

RCT $\Rightarrow$ ATE: Binary $X_i \in \{0, 1\}$ randomly assigned: $\beta_1 = \mathbb{E}[Y_i | X_i=1] - \mathbb{E}[Y_i | X_i=0]$ = average treatment effect.

Binary X (general): Any binary X : $\beta_0 = \mathbb{E}[Y | X=0]$, $\beta_1 = \mathbb{E}[Y | X=1] - \mathbb{E}[Y | X=0]$ – a *difference in means*, not a slope.

Selection/survivorship recipe: (1) name the unobserved factor sitting in u (ability, health, motivation); (2) explain why it drives *both* selection into the sample/into X *and* the outcome Y ; (3) state the direction of bias. Hospital survey: sicker people select into hospital $\Rightarrow \text{Cov}(\text{hospital}, u) \neq 0$. Wald bombers: only planes that *returned* are observed $\Rightarrow$ reinforce the *least-damaged* areas, not the most.

RCT vs. observational: RCT: random X -assignment $\Rightarrow$ LSA1 typically holds *by design*. Observational data (education, HSR lines): X not randomly handed out $\Rightarrow$ LSA1 needs case-by-case justification.

Return-to-education example: wage = $\beta_0 + \beta_1 \text{educ} + u$, $u \ni$ innate ability: LSA1 needs $\mathbb{E}[\text{abil}|\text{educ}]$ constant across education levels – fails if higher-ability people self-select into more schooling.

LSA2/LSA3 violation examples: LSA2 fails: time series (firm revenue over 30 yrs, autocorrelated), non-random sampling (interviewing at city parks at 10am misses employed people). LSA3 fails: a handful of extreme outliers dominate the squared-error objective.

4. Sampling & Fit

SRS $\Rightarrow$ i.i.d.: Simple random sampling: every population member equally likely drawn $\Rightarrow$ independence across i ; all drawn from the same population $\Rightarrow$ identical distribution. Time series is the standard counterexample (autocorrelated).

TSS/ESS/SSR: $Y_i = \hat{Y}_i + \hat{u}_i$. TSS = $\sum_i (Y_i - \bar{Y})^2$ (total); ESS = $\sum_i (\hat{Y}_i - \bar{Y})^2$ (explained); SSR = $\sum_i \hat{u}_i^2$ (residual). TSS = ESS + SSR (cross term $2 \sum_i \hat{u}_i \hat{Y}_i - 2 \bar{Y} \sum_i \hat{u}_i = 0$). Never write “SSE”.

R^2 : $R^2 = \text{ESS}/\text{TSS} = 1 - \text{SSR}/\text{TSS} \in [0, 1]$ since $\text{SSR} \geq 0$; $R^2 = [\text{Corr}(X, Y)]^2$. $\hat{\beta}_1 = 0 \Rightarrow \hat{Y}_i = \hat{\beta}_0 = \bar{Y} \Rightarrow \text{ESS} = 0 \Rightarrow R^2 = 0$.

Corr(X, Y) algebra: $\text{Corr}(X, Y) = s_{XY}/(s_X s_Y)$; since $\hat{\beta}_1 = s_{XY}/s_X^2$, $\hat{\beta}_1 = \text{Corr}(X, Y) \cdot (s_Y/s_X)$ – links the slope directly to the correlation coefficient and the two sample SDs.

Worked TSS/ESS/SSR: $n = 4$: $X = (1, 2, 3, 4)$, $Y = (3, 3, 5, 9)$, OLS fit $\hat{Y} = 2X$ (so $\hat{\beta}_0 = 0, \hat{\beta}_1 = 2$): $\bar{Y} = 5$, $\hat{u} = (1, -1, -1, 1)$. TSS = $4+4+0+16 = 24$; ESS = $9+1+1+9 = 20$; SSR = $1+1+1+1 = 4$; ESS + SSR = 24 = TSS; $R^2 = 20/24 = 0.833$.

Mean of $\hat{Y}$: $\frac{1}{n} \sum_i \hat{Y}_i = \bar{Y}$ (from $\sum_i \hat{u}_i = 0$, or directly from $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$).

$\bar{Y}$ mean & variance: Y_i i.i.d. (μ, σ^2) : $\mathbb{E}[\bar{Y}] = \mu$ (needs only identical dist.); $\text{Var}[\bar{Y}] = \sigma^2/n$ (needs independence: cross-covariances vanish). $\bar{Y} \stackrel{\text{approx}}{\sim} N(\mu, \sigma^2/n)$ for $n > 100$ (CLT).

Var($\bar{Y}$) derivation: $\text{Var}(\bar{Y}) = \text{Var}(\frac{1}{n} \sum_i Y_i) = \frac{1}{n^2} \sum_i \text{Var}(Y_i) + \frac{1}{n^2} \sum_{i \neq j} \text{Cov}(Y_i, Y_j) = \frac{1}{n^2} (n \sigma^2) + 0 = \sigma^2/n$ (indep. kills the cross terms).

Sample variance s^2 : $s^2 = \frac{1}{n-1} \sum_i (Y_i - \bar{Y})^2$ estimates σ^2 ($n-1$ divisor

corrects for using $\bar{Y}$ instead of μ); same idea underlies $\hat{\sigma}_{\hat{\beta}_1}^2$ in Sec. 6.

CAPM: $R - R_f = \alpha + \beta(R_m - R_f) + u$. Strict CAPM: $\alpha = 0$; $\hat{\alpha} > 0$ significant = “beats the market”. $\beta < 1$: less systematic risk & lower expected excess return than market; $\beta > 1$: more risk & higher expected excess return.

Diff. of two indep. means: $\bar{Y}_1, \bar{Y}_2$ from independent samples, sizes N_1, N_2 : $\mathbb{E}[\bar{Y}_1 - \bar{Y}_2] = \mu_1 - \mu_2$; $\text{Cov}(\bar{Y}_1, \bar{Y}_2) = 0$; $\text{Var}(\bar{Y}_1 - \bar{Y}_2) = \sigma_1^2/N_1 + \sigma_2^2/N_2$.

Hospital survey numbers: Hospital group: $n = 7,774$, mean health 3.21; non-hospital: $n = 90,049$, mean 3.93. Naive gap 0.72 is *not* the causal effect of hospitalisation – self-selection, not treatment, drives most of it.

Weight/height reference: $n = 100$: $\widehat{\text{Weight}} = -79.24 + 4.16 \text{Height}$ (lb, in). At Height= 68: $\hat{Y} = 203.6$; a 2-inch growth spurt $\Rightarrow \Delta \hat{Y} = 8.32$ lb.

Low R^2 is not a red flag: TestScore on STR: $R^2 = 0.051$ – small, but says nothing about whether $\hat{\beta}_1$ is a valid *causal* estimate; that is an LSA1 question, not an R^2 question.

5. Unbiasedness, Efficiency & Dummies

Linear estimator of μ : $\hat{\mu} = \sum_i a_i Y_i$, Y_i i.i.d. (μ, σ^2) : $\mathbb{E}[\hat{\mu}] = \mu \sum_i a_i$, so $\hat{\mu}$ unbiased $\iff \sum_i a_i = 1$.

Variance of $\hat{\mu}$: $\text{Var}(\hat{\mu}) = \sigma^2 \sum_i a_i^2$ (independence kills cross terms). With $\sum_i a_i = 1$: $\sum_i a_i^2 = \sum_i (a_i - 1/n)^2 + 1/n \geq 1/n$, minimised at $a_i = 1/n \forall i$. $\bar{Y}$ is **BLUE**: Among all linear unbiased $\hat{\mu} = \sum_i a_i Y_i$: $\text{Var}(\bar{Y}) = \sigma^2/n \leq \text{Var}(\hat{\mu})$ – Best Linear Unbiased Estimator of μ .

$\hat{\beta}_1$ **as a weighted sum:** $\hat{\beta}_1 = \sum_i w_i Y_i$, $w_i = \frac{X_i - \bar{X}}{\sum_j (X_j - \bar{X})^2}$ ($\sum_i w_i = 0$,

$\sum_i w_i X_i = 1$) – the OLS weights, used as the starting point of the Gauss–Markov proof.

OLS unbiasedness proof sketch: Substitute $Y_i = \beta_0 + \beta_1 X_i + u_i$ into $\hat{\beta}_1 = \sum_i w_i Y_i$: $\hat{\beta}_1 = \beta_1 \underbrace{\sum_i w_i X_i}_{=1} + \beta_0 \underbrace{\sum_i w_i}_{=0} + \sum_i w_i u_i = \beta_1 + \sum_i w_i u_i$.

Then $\mathbb{E}[\hat{\beta}_1 | X] = \beta_1 + \sum_i w_i \mathbb{E}[u_i | X_i] = \beta_1$ by LSA1.

Var($\hat{\beta}_1$) population formula: Under homoskedasticity: $\text{Var}(\hat{\beta}_1 | X) =$

$\sigma_u^2 \sum_i w_i^2 = \frac{\sigma_u^2}{\sum_i (X_i - \bar{X})^2}$ – matches Sec. 6: larger spread of X or smaller $\sigma_u^2 \Rightarrow$ smaller $\text{Var}(\hat{\beta}_1)$.

Gauss–Markov proof sketch: Any alternative linear unbiased $\tilde{\beta}_1 = \sum_i (w_i + c_i) Y_i$ needs $\sum_i c_i = 0, \sum_i c_i X_i = 0$ (for unbiasedness); then $\text{Var}(\tilde{\beta}_1) = \text{Var}(\hat{\beta}_1) + \sigma_u^2 \sum_i c_i^2 \geq \text{Var}(\hat{\beta}_1)$, equality iff $c_i = 0 \forall i$.

Gauss–Markov: Under LSA1–3 *plus* homoskedasticity $\text{Var}(u_i | X_i) = \sigma_u^2$ (constant): $\hat{\beta}_0, \hat{\beta}_1$ are BLUE – $\text{Var}(\hat{\beta}_0) \leq \text{Var}(\tilde{\beta}_0)$, $\text{Var}(\hat{\beta}_1) \leq \text{Var}(\tilde{\beta}_1)$ for *any* linear unbiased $\tilde{\beta}_0, \tilde{\beta}_1$. Without homoskedasticity, OLS stays unbiased but need not be most efficient.

Binary regressor: $Y_i = \beta_0 + \beta_1 D_i + u_i$, $D_i \in \{0, 1\}$: $\hat{\beta}_0 = \bar{Y}_{D=0}$, $\hat{\beta}_0 + \hat{\beta}_1 = \bar{Y}_{D=1}$, so $\hat{\beta}_1 = \bar{Y}_{D=1} - \bar{Y}_{D=0}$ (sample group-mean gap).

Common slope, diff. intercepts: $Y_i = \beta_0 + \delta_0 D_i + \beta_1 X_i + u_i$; $D_i=0$ line $\beta_0 + \beta_1 X_i$; $D_i=1$ line $(\beta_0 + \delta_0) + \beta_1 X_i$ – parallel lines, vertical gap δ_0 .

Multi-dummy & base group: k categories $\Rightarrow$ include $k-1$ dummies + intercept. Omitted category = **base group**, absorbed into β_0 ; each dummy’s coefficient reads “vs. base group, holding other regressors fixed.”

Dummy variable trap: All k category dummies *and* an intercept together $\Rightarrow$ dummies sum to 1 for every i , an exact linear combo of the constant regressor $\Rightarrow$ perfect multicollinearity, no unique OLS solution. Fix: drop one category.

Simulation intuition: $n = 10$ draws from $N(2, 1)$, 20 sims: $Y_1 \in [-0.64, 4.27]$ vs. $\bar{Y} \in [1.16, 2.58]$ – $\bar{Y}$ closer to $\mu = 2$ in 16/20 sims, il-

lustrating $\text{Var}(Y_1) = \sigma^2 > \text{Var}(\bar{Y}) = \sigma^2/n$.
Law School model: $\text{salary}_i = \beta_0 + \beta_1 \text{top10}_i + \beta_2 \text{r11.25}_i + \beta_3 \text{r26.50}_i + \beta_4 \text{GPA}_i + u_i$; base group = ranked below 50th. Top-10 line: $(\beta_0 + \beta_1) + \beta_4 \text{GPA}_i$; each dummy coefficient = gap vs. base group, same GPA.
Hypothesis-test setup (L5): $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$; reject if $|\bar{Y} - \mu_0| > c$. Type I = reject true H_0 ; Type II = fail to reject false H_0 ; significance level = $P(\text{Type I} | H_0 \text{ true})$. Same logic sets up $H_0 : \beta_1 = 0$ (STR has no effect); made operational via $\text{SE}(\hat{\beta}_1)$ in Sec. 6.
Common-slope example: California regions: $D_i = 1$ if Region A, 0 if Region B, added alongside STR; δ_0 = region gap in test scores *not* explained by STR (a parallel-line shift).

6. Sampling Distribution of $\hat{\beta}_1$ (L6 pp.1–3)

$\hat{\beta}_1$ is a random variable: $\hat{\beta}_1$ is computed from a random sample, so it varies from sample to sample (a sampling distribution, not a fixed number); this is the object Sec. 6 studies, distinct from the within-sample spread of u_i .
Large-sample normality: Under LSA1–3, n large: $\hat{\beta}_1 \approx N(\beta_1, \sigma_{\hat{\beta}_1}^2)$. Standardised version: $(\hat{\beta}_1 - \beta_1)/\sigma_{\hat{\beta}_1} \approx N(0, 1)$.
 $\sigma_{\hat{\beta}_1}^2$ & $\text{SE}(\hat{\beta}_1)$: A variance estimator $\hat{\sigma}_{\hat{\beta}_1}^2$ (built from $\hat{u}_i$ and $(X_i - \bar{X})^2$) exists – formula need not be recited. $\text{SE}(\hat{\beta}_1) = \sqrt{\hat{\sigma}_{\hat{\beta}_1}^2}$ estimates the spread/dispersion of $\hat{\beta}_1$ ’s sampling distribution about β_1 .
What makes $\text{SE}(\hat{\beta}_1)$ small: (qualitative) larger n ; larger spread (variance) of X_i – more leverage to trace the slope; smaller error variance σ_u^2 – points hug the line more tightly.
Standardised estimator: In large samples $\text{SE}(\hat{\beta}_1) \approx \sigma_{\hat{\beta}_1}$, so $\frac{\hat{\beta}_1 - \beta_1}{\text{SE}(\hat{\beta}_1)} \approx N(0, 1)$ – same standardisation recipe as any $N(\mu, \sigma^2)$ variable.
Standardisation recipe: $Z = \frac{\text{value} - \text{mean}}{\text{sd}}$ turns any $N(\mu, \sigma^2)$ variable into $N(0, 1)$. E.g. $\hat{\beta}_1 = 2.28, \beta_1 = 2, \sigma_{\hat{\beta}_1} = 0.5 \Rightarrow z = (2.28 - 2)/0.5 = 0.56$.
 $P(\hat{\beta}_1 > c)$ **recipe:** Standardise: $P(\hat{\beta}_1 > c) = P\left(Z > \frac{c - \beta_1}{\sigma_{\hat{\beta}_1}}\right) = 1 - \Phi\left(\frac{c - \beta_1}{\sigma_{\hat{\beta}_1}}\right)$; read Φ off the table (use $\text{SE}(\hat{\beta}_1)$ in place of $\sigma_{\hat{\beta}_1}$ if only the estimate is known).
Symmetric-interval probability: $P(|\hat{\beta}_1 - \beta_1| < k \sigma_{\hat{\beta}_1}) = P(-k < Z < k) = 2\Phi(k) - 1$; e.g. $k = 1.96 \Rightarrow 0.95, k = 1 \Rightarrow 0.6827$. Read $\Phi(k)$ off the table.
Worked example: $\beta_1 = 2, \sigma_{\hat{\beta}_1} = 0.5: P(\hat{\beta}_1 > 2.98) = P\left(Z > \frac{2.98 - 2}{0.5}\right) = P(Z > 1.96) = 1 - \Phi(1.96) = 1 - .9750 = .0250$.
Big-picture recap: (1) model $Y_i = \beta_0 + \beta_1 X_i + u_i$ & target β_1 (Sec. 2);

(2) estimator $\hat{\beta}_1$ (Sec. 2); (3) unbiasedness & efficiency under LSA1–3 (Sec. 5); (4) large-sample distribution $\hat{\beta}_1 \approx N(\beta_1, \sigma_{\hat{\beta}_1}^2)$ and $\text{SE}(\hat{\beta}_1)$ (Sec. 6); (5) standardise & read $\Phi(z)$ for probability questions.

Reference Numbers (empirical examples)

Dataset (n)	Fitted line	SE	Used for
California TestScore/STR (420) CEO salary/roe (209)	698.9 – 2.28STR 963.2 + 18.5 roe	.52 6.83	fit, prediction LSA1, residual, SE
Weight/Height (100) Hospital survey means (7,774/90,049)	–79.2 + 4.16Ht 3.21 vs. 3.93	– –	prediction group means, LSA1

Notation glossary: β : true param; b : candidate in minimization; $\hat{\beta}$: OLS estimate. u : true error; $\hat{u}$: residual. Y : observed; $\hat{Y}$: fitted. μ, σ^2 : population mean/var; $\bar{Y}, s^2$: sample mean/var. w_i : OLS weight on Y_i ; $\sigma_{\hat{\beta}_1}^2, \hat{\sigma}_{\hat{\beta}_1}^2$: (estimated) variance of $\hat{\beta}_1$ ’s sampling distribution; D_i : dummy/indicator regressor.

Standard Normal $\Phi(z)$ Table

z	.500	1.000	1.280	1.500	1.645
$\Phi(z)$	.6915	.8413	.8997	.9332	.9500
z	1.960	2.000	2.500	2.580	3.000
$\Phi(z)$	.9750	.9772	.9938	.9951	.9987

Using this table: $\Phi(z) = P(Z \leq z)$ for $Z \sim N(0, 1)$; by symmetry $\Phi(-z) = 1 - \Phi(z)$. Plug $z = (\text{value} - \text{mean})/\text{sd}$ from the standardisation recipe, then read $\Phi(z)$ off here – not a test critical value, just a probability lookup.

Triggers & Traps

“causal effect”: needs LSA1 $\mathbb{E}[u | X] = 0$; ask what is in u and whether it correlates with X .
“binary X ”: $\beta_1 = \mathbb{E}[Y | X=1] - \mathbb{E}[Y | X=0]$, a difference in means, not a slope.
“interpret a dummy coefficient”: read it as “vs. the omitted/base group, holding other regressors fixed” (Sec. 5); never as a per-unit slope like a continuous X ’s coefficient.
“show unbiased”: compute $\mathbb{E}[\sum_i a_i Y_i] = \mu \sum_i a_i$ and require $\sum_i a_i = 1$.
“ $R^2 \leq 1$ ”: chase the identity chain: $\text{TSS} = \sum_i [\hat{u}_i + (\hat{Y}_i - \bar{Y})]^2 \Rightarrow \text{TSS} = \text{ESS} + \text{SSR}$ (cross term = 0 via $\sum_i \hat{u}_i \hat{Y}_i = 0, \sum_i \hat{u}_i = 0) \Rightarrow \text{SSR} \geq 0 \Rightarrow R^2 \leq 1$.

“probability that the estimate exceeds/is below c ”: standardise with $\sigma_{\hat{\beta}_1}$ (or $\text{SE}(\hat{\beta}_1)$ if that’s given): $z = (c - \beta_1)/\sigma_{\hat{\beta}_1}$, then use $\Phi(z)$ (upper tail = $1 - \Phi(z)$).
“why is $\hat{\beta}_1$ random?”: it depends on *which sample* is drawn – a different sample gives a different $\hat{\beta}_1$; that variability across hypothetical repeated samples is its sampling distribution (Sec. 6).
“ E and Var of a difference of two independent means”: $\mathbb{E}[\bar{Y}_1 - \bar{Y}_2] = \mu_1 - \mu_2$; independence $\Rightarrow \text{Cov}(\bar{Y}_1, \bar{Y}_2) = 0 \Rightarrow \text{Var}(\bar{Y}_1 - \bar{Y}_2) = \sigma_1^2/N_1 + \sigma_2^2/N_2$ (Sec. 4).
“ H_0/H_1 for a mean” (L5): set up $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$; report Type I/Type II and the significance level $P(\text{Type I} | H_0)$ – this framework is in scope, the t -statistic machinery of L6 pp.4+ is not.
“efficient” / “smallest variance”: compare only among *linear, unbiased* estimators (Gauss–Markov); never compare variance across a biased vs. unbiased pair to declare a winner.
“find $\text{Var}(\hat{\beta}_1)$ ” / “what does $\text{SE}(\hat{\beta}_1)$ estimate”: population formula $\text{Var}(\hat{\beta}_1 | X) = \sigma_u^2 / \sum_i (X_i - \bar{X})^2$ (homoskedastic case); $\text{SE}(\hat{\beta}_1)$ is its sample estimate – same small/large drivers as Sec. 6 (larger n or spread of X , smaller σ_u^2).
“ $\bar{Y}$ is approximately normal” / CLT: Y_i i.i.d. $(\mu, \sigma^2) \Rightarrow \bar{Y} \stackrel{\text{approx}}{\sim} N(\mu, \sigma^2/n)$ for $n > 100$; the same CLT logic gives $\hat{\beta}_1 \approx N(\beta_1, \sigma_{\hat{\beta}_1}^2)$ in Sec. 6.
“sample selects itself” / “survivorship”: an LSA1 failure: name the omitted factor in u driving both selection and Y .
Trap: u_i **vs.** $\hat{u}_i$: only *residuals* $\hat{u}_i$ provably sum to zero (from the FOCs); the true error u_i need not.
Trap: $\mathbb{E}[u] = 0$ **is not enough:** LSA1 needs $\mathbb{E}[u | X] = 0$ at *every* X , not just the unconditional mean; $u = X^2 - 1$ is the standard counterexample.
Trap: k **dummies + intercept:** dummy variable trap – always use $k - 1$ dummies for k categories.
“5-step big picture” / “big picture framework”: model + parameter $\rightarrow$ estimator $\hat{\beta}_1 \rightarrow$ its (large-sample) distribution $\rightarrow \text{SE}(\hat{\beta}_1) \rightarrow$ standardise & read $\Phi(z)$.
Trap: quoting $\hat{\beta}_0$ at $X = 0$: only a valid prediction within the observed range of X ; extrapolating to $X = 0$ outside the sample range is unreliable.
“population line” vs. “OLS line”: population regression line $\beta_0 + \beta_1 X$ is unknown/true; OLS line $\hat{\beta}_0 + \hat{\beta}_1 X$ is estimated from the sample – always keep the two distinct in an answer.
“sampling distribution” vs. “distribution of u ”: sampling distribution of $\hat{\beta}_1$ = variability across hypothetical repeated samples; distribution of u_i = spread *within* one sample. Different objects, both used in Sec. 4–6.
“holding other regressors fixed”: multi-dummy or CAPM-style controls: read a coefficient as the gap/effect *at fixed values of the other regressors* (Sec. 5 Law School model, GPA held fixed across rank dummies).
Trap: **dropping the sign:** always keep the sign of $\hat{\beta}_1$ (and of any standardised z) in your final answer – the sign carries the economic/probability direction.